

IDS-Twin: Digital Twin-Based Intrusion Detection Systems for Wireless Networks

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Abstract—The rapid expansion of IoT devices has greatly enhanced wireless connectivity but also increased exposure to sophisticated and evolving cyber threats. Recent intrusion detection systems (IDS) based on Artificial Intelligence (AI) often rely on offline training and static models, limiting their ability to adapt in dynamic environments where device behavior and attack patterns continuously change. To address these challenges, we propose IDS-Twin, a Digital Twin (DT)-based intrusion detection framework designed for wireless IoT. IDS-Twin incorporates a self-adaptive anomaly detection model powered by contrastive learning, enabling real-time learning from streaming traffic with minimal labeled data. The DT component is implemented as a lightweight software replay engine that emulates real-time arrival by sequentially streaming flows in timestamp order, enabling the system to continuously detect and adapt to new or evolving attacks. Experimental validation on the UNSW-NB15 and CIC-IoT-2023 datasets demonstrates the superior performance and adaptability of IDS-Twin, outperforming state-of-the-art IDS solutions in detecting both known and unknown attacks in wireless networks.

Index Terms—Intrusion detection systems, cybersecurity, digital twin, Internet of Things, artificial intelligence, contrastive learning.

I. INTRODUCTION

INTRUSION detection systems (IDS) has become a critical component of modern network security architectures, especially given the increasing complexity and frequency of cyberattacks [1], [2]. The rapid advancement of wireless communication and Internet of Things (IoT) systems necessitates the development of real-time, intelligent, and adaptive IDS. In this context, machine learning (ML)-based IDS have demonstrated significant potential [3], [4], particularly in learning complex traffic patterns to identify intrusions. These methods enable rapid deployment and achieve high accuracy in anomaly detection and intrusion prevention, particularly within IoT security. For example, the authors in [5] performed a comprehensive evaluation comparing ensemble methods to individual classifiers. The study in [6] introduced a lightweight IDS based on supervised learning to detect cyberattacks and anomalies in resource-constrained IoT environments. A contrastive learning-based approach was proposed in [7] as a novel method to enhance ML classifiers and improve

the accuracy of intrusion detection. In addition, the study in [8] proposed a novel online intrusion detection framework based on contrastive learning. It incorporates an autonomous anomaly detection module and a label-free online mechanism for continuous adaptation. In the same context, the work in [9] introduces a Digital Twin (DT)-based intrusion detection approach for industrial control systems, where a stacked ensemble classifier is constructed using offline evaluations of eight supervised ML models to support real-time detection. Although ML has shown great promise in capturing complex traffic patterns for intrusion detection, its performance in real-world IoT deployments remains limited by the dynamic nature of modern networks [10]. This limitation is largely driven by the explosive growth of heterogeneous IoT devices and diverse application protocols, which produce highly variable and unpredictable traffic behaviors. Such variability introduces significant intra-class variation, making it difficult for ML models to maintain consistent representations across time. ML-based IDS often struggle to identify zero-day attacks because they rely heavily on historical labeled data and lack the adaptability to detect previously unseen or evolving threat behaviors [11], [12]. This limitation is further intensified by the behavioral convergence between benign and malicious traffic, which reduces class separability and degrades detection accuracy. In addition to these algorithmic limitations, deploying ML-based IDS in real-time environments can be resource-intensive, as it frequently requires specialized hardware and supporting infrastructure. These limitations highlight the need for a new intrusion detection paradigm, one that can learn robust, discriminative representations from unlabeled and continuously evolving traffic data while preserving contextual awareness of system behavior.

To address these challenges, we propose IDS-Twin, a DT-based IDS that integrates the DT and contrastive learning. The DT serves as a virtual environment that replays traffic flows in timestamp order, allowing the IDS to simulate real-time IoT network behavior and adapt to evolving conditions. Meanwhile, contrastive learning helps IDS-Twin extract meaningful representations by pulling similar flows together and pushing dissimilar ones apart. It improves detection accuracy and enables the system to recognize unseen (zero-day) attacks during online operation. This synergy enables dynamic traffic analysis, timely anomaly detection, and adaptive responses to both known and previously unseen threats. In this letter, the DT is implemented as a software-defined replay engine. Instead of interfacing with physical devices, it sequentially replays pre-collected IoT flows in their original timestamp order. This emulates real-time arrival while ensuring controlled, reproducible evaluation. Thus, the DT provides a lightweight virtual environment that mirrors temporal

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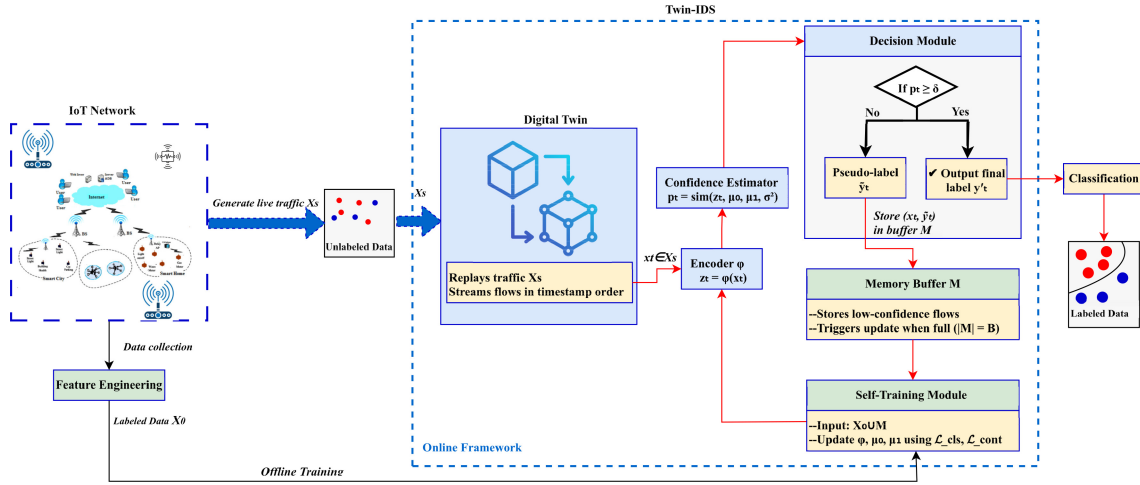


Fig. 1. End-to-end architecture of IDS-Twin for adaptive streaming-based intrusion detection.

dynamics of network traffic without requiring physical-system coupling. This setup enables dynamic emulation of system behavior without requiring physical coupling, similar to several prior DT-based security frameworks [13]. In this letter, we propose IDS-Twin, a novel DT-based intrusion detection framework that combines contrastive and classification losses to enable adaptive online learning from streaming wireless IoT traffic. The system incorporates a Gaussian confidence-based decision mechanism and a buffer-driven self-training module, allowing it to update class prototypes effectively without requiring explicit drift detection. We also develop a lightweight Python-based DT simulation for time-aware traffic replay and evaluate IDS-Twin on the UNSW-NB15 [14] and CIC-IoT-2023 [15] datasets, demonstrating its superiority over state-of-the-art (SOTA) IDS baselines.

II. PROBLEM FORMULATION

We consider a wireless IoT environment in which a DT serves as a virtual replica, streaming traffic data in real-time to simulate the behavior of edge devices and network nodes. The objective is to enable online, adaptive intrusion detection using a self-evolving learning system. Let $\mathcal{X} = \{x_1, x_2, \dots, x_n\}$ be a sequence of d -dimensional flow feature vectors arriving in temporal order, with partial label supervision $\mathcal{Y}_0 \subset \mathcal{Y}$. The goal is to learn a binary classifier $f_\theta : \mathbb{R}^d \rightarrow \{0, 1\}$ under three constraints: streaming data, limited labels, and non-stationary distributions. We seek to minimize the expected classification loss over a time-varying distribution $\mathcal{D}_t$:

$$\min_{\theta} \mathbb{E}_{(x,y) \sim \mathcal{D}_t} [\ell(f_\theta(x), y)] \quad (1)$$

To detect zero-day attacks, samples not drawn from the known support of labeled malicious traffic, we define a latent representation function $\varphi : \mathbb{R}^d \rightarrow \mathbb{R}^z$ and compute class prototypes ω_0 and ω_1 . A sample x_i is flagged as a potential zero-day if its cosine similarity to both prototypes is below a margin γ :

$$\max\{\cos(\varphi(x_i), \omega_0), \cos(\varphi(x_i), \omega_1)\} < \gamma \quad (2)$$

This formulation enables the system to identify novel threats without requiring additional labels, supporting adaptive detection in evolving wireless network.

The DT state at time t is denoted as $s_t = (G_t, \Pi_t, R_t, C_t, F_t)$, where G_t is topology, Π_t protocol context, R_t device roles, C_t control logic, and F_t flow telemetry. In this letter, only F_t is explicitly instantiated from datasets, while the remaining components are abstracted but conceptually define the DT's fidelity boundary.

III. IDS-TWIN DESIGN

Fig. 1 illustrates the end-to-end architecture of IDS-Twin, a lightweight and adaptive intrusion detection framework designed for streaming-based IoT environments. The system consists of three key modules: (i) a supervised contrastive representation learner (Encoder ϕ), (ii) a Gaussian-based confidence estimator with decision module, and (iii) a self-training module triggered by a memory buffer. A DT receives the unlabeled flow stream $X_s = x_t, t = 1^\infty$ from IoT devices and replays each flow x_t sequentially in timestamp order for real-time analysis. In this letter, the DT is implemented as a software-defined replay engine without direct interfaces to sensors or devices. This abstraction provides a controlled and reproducible evaluation setting for IDS-Twin under streaming conditions. Each flow x_t is first encoded, and its confidence score p_t is computed using the Gaussian probability model. Confident predictions ($p_t \geq \delta$) yield final labels y'_t , while uncertain samples ($p_t < \delta$) are assigned pseudo-labels $\tilde{y}_t$ via cosine similarity, which are stored for buffer-driven self-adaptation. Offline initialization is performed on labeled data X_0 to bootstrap the encoder. *Algorithm 1* summarizes the IDS-Twin workflow.

Offline initialization phase. In the initialization stage, a labeled dataset $X_0 = \{(x_i, y_i)\}_{i=1}^n$ is used to train an encoder-decoder autoencoder (ϕ, ψ) . The encoder $\phi : \mathbb{R}^d \rightarrow \mathbb{R}^m$ is implemented as a shallow Multilayer Perceptron (MLP) with two fully connected layers and ReLU activations, mapping raw flow samples x_i into latent embeddings $z_i = \phi(x_i)$. The decoder ψ , structured as a mirror of the encoder, is used only during pretraining to ensure stable feature extraction and is

Algorithm 1: IDS-Twin

Input: Labeled dataset X_0 , streaming flows $X_s = \{x_t\}_{t=1}^\infty$, buffer size B , confidence threshold δ

Output: Predicted label y'_t or pseudo-label $\tilde{y}_t$ for each $x_t \in X_s$

Offline Initialization Phase: // One-time training on X_0
 Train encoder-decoder (ϕ, ψ) on X_0 using total loss $\mathcal{L}_{\text{total}}$;
 Discard decoder ψ after training ;
 Compute embeddings $z_i = \phi(x_i)$ for all $x_i \in X_0$;
 Compute class centroids μ_0, μ_1 and pooled variance σ^2 ;
 Initialize memory buffer $\mathcal{M} \leftarrow \emptyset$;

Online Adaptive Detection Phase: // Streaming classification and self-adaptation
for each incoming flow $x_t \in X_s$ **do**
 Encode: $z_t = \phi(x_t)$;
 Compute cosine similarities: $s_t^k = \cos(z_t, \mu_k)$ for $k \in \{0, 1\}$;
 Compute confidence score p_t using (4) ;
 if $p_t \geq \delta$ // Confident prediction
 then
 Predict: y'_t based on p_t ;
 else
 // Low confidence $\rightarrow$ Drift indicator
 Assign pseudo-label: $\tilde{y}_t = \arg \max_k s_t^k$;
 Store $(x_t, \tilde{y}_t)$ in buffer $\mathcal{M}$;
 end
 if $|\mathcal{M}| \geq B$ // Trigger self-adaptation
 then
 Augment training set: $X \leftarrow X_0 \cup \mathcal{M}$;
 Recompute embeddings: $z_i = \phi(x_i)$ for $x_i \in X$;
 Retrain encoder ϕ on X using $\mathcal{L}_{\text{total}}$;
 Update μ_0, μ_1 , and σ^2 from new embeddings ;
 Reset buffer: $\mathcal{M} \leftarrow \emptyset$;
 end
end

discarded afterward. A joint objective combines supervised contrastive learning with probabilistic classification to guide the encoder toward class-consistent, semantically separable representations, with the contrastive loss $\mathcal{L}_{\text{cont}}$ defined as:

$$\mathcal{L}_{\text{cont}}(z_i, z_j) = -\log \left(\frac{\exp\left(\frac{\text{sim}(z_i, z_j)}{\tau}\right)}{\sum_{k \neq i} \exp\left(\frac{\text{sim}(z_i, z_k)}{\tau}\right)} \right) \quad (3)$$

where z_j is a positive sample from the same class as z_i , $\text{sim}(z_i, z_j) = \frac{z_i^\top z_j}{\|z_i\| \|z_j\|}$ denotes cosine similarity, and τ is the temperature parameter. This loss encourages embeddings of similar samples to cluster while separating dissimilar ones, enhancing generalization under distribution shift. To incorporate label supervision and support probabilistic inference, a Gaussian confidence model is introduced. After training the encoder, class centroids μ_0 and μ_1 (representing benign and malicious classes, respectively) and pooled variance σ^2 are computed over the embeddings of X_0 . For a new embedding z_t , the probability of being malicious is estimated as:

$$p_t = \frac{\exp\left(-\frac{\|z_t - \mu_1\|^2}{2\sigma^2}\right)}{\exp\left(-\frac{\|z_t - \mu_0\|^2}{2\sigma^2}\right) + \exp\left(-\frac{\|z_t - \mu_1\|^2}{2\sigma^2}\right)} \quad (4)$$

This probability is used to compute the classification loss $\mathcal{L}_{\text{cls}}$:

$$\mathcal{L}_{\text{cls}} = -y_t \log(p_t) - (1 - y_t) \log(1 - p_t) \quad (5)$$

which penalizes incorrect predictions based on soft confidence values rather than hard labels. The total training loss is a weighted combination of the two objectives:

$$\mathcal{L}_{\text{total}} = \alpha \cdot \mathcal{L}_{\text{cls}} + \beta \cdot \mathcal{L}_{\text{cont}} \quad (6)$$

where $\alpha + \beta = 1$ and both $\alpha, \beta \in [0, 1]$ control the trade-off between classification performance and representation robustness. The encoder ϕ is optimized using gradient-based training on X_0 to minimize $\mathcal{L}_{\text{total}}$. After training, the decoder ψ is discarded. The centroids μ_0, μ_1 , and variance σ^2 are retained for online detection, and buffer $\mathcal{M}$ is initialized to collect low-confidence samples during inference.

Online detection phase. During deployment, IDS-Twin receives an incoming stream of unlabeled network flows denoted by $X_s = \{x_t\}_{t=1}^\infty$, where $x_t \in \mathbb{R}^d$ represents the t -th flow feature vector. Each flow x_t is first passed through the trained encoder ϕ to produce a latent representation $z_t = \phi(x_t)$. The cosine similarities $s_t^k = \cos(z_t, \mu_k)$ are then computed with respect to the stored class centroids μ_k for $k \in \{0, 1\}$, and the probabilistic confidence score p_t is estimated using the Gaussian model in (4). A decision rule is applied based on a predefined confidence threshold $\delta \in (0, 1)$. If $p_t \geq \delta$, the sample is considered confidently classifiable, and the final prediction y'_t is determined directly from the Gaussian probability model in (4). If $p_t < \delta$, the system does not trust this probability and instead assigns a pseudo-label $\tilde{y}_t$ using cosine similarity ($\arg \max_k s_t^k$), which is stored in the buffer for later self-adaptation. The pair $(x_t, \tilde{y}_t)$ is then appended to the memory buffer $\mathcal{M}$. Confidence-guided selection lets IDS-Twin defer uncertain predictions, gather useful samples for adaptation, and maintain robustness to shifts and zero-day attacks without requiring immediate ground-truth labels. In addition, Detection outputs can be exported as alerts to monitoring platforms or as policy updates to enforcement points such as gateways or host firewalls. With an inference latency of ≈ 6.8 ms per flow and update cycles every 800–1000 flows, the detection-to-alert process operates within sub-second timescales.

Self-adaptation phase. When the memory buffer $\mathcal{M}$ accumulates B low-confidence pseudo-labeled samples, the self-adaptation phase is triggered. At this point, the training set is augmented as $X \leftarrow X_0 \cup \mathcal{M}$, and the encoder ϕ is retrained over the combined data using the same contrastive-classification objective $\mathcal{L}_{\text{total}}$ defined during the offline phase. For all $x \in X$, new embeddings $z = \phi(x)$ are computed. The class centroids μ_0, μ_1 and pooled variance σ^2 are then updated using these new representations. Once the update is completed, the memory buffer $\mathcal{M}$ is cleared to prepare for the next adaptation cycle. This asynchronous periodic retraining reduces the cost of continuous updates while allowing the system to realign with evolving traffic distributions and emerging threats.

Implicit drift awareness. Unlike conventional concept drift detection methods that rely on explicit statistical alarms, IDS-Twin adopts a lightweight and implicit mechanism based on confidence degradation. A persistent occurrence of streaming samples $x_t \in X_s$ with confidence scores $p_t < \delta$ is treated as an indicator of distributional drift. These low-confidence flows are stored in the memory buffer $\mathcal{M}$ and subsequently

used to retrain the encoder ϕ during the self-adaptation phase. IDS-Twin implicitly adapts to distribution shifts by realigning decision boundaries without explicit drift detectors, avoiding modules like Drift Detection Method (DDM) or Adaptive Windowing (ADWIN) and improving efficiency for dynamic IoT.

IV. PERFORMANCE EVALUATION

A. Experimental Setup

Dataset setup, baselines, and evaluation metrics. To evaluate IDS-Twin, we used the UNSW-NB15 [14] and CIC-IoT-2023 [15] datasets, both containing diverse traffic patterns with benign and malicious flows (e.g., DDoS, probing, Web-based attacks). Preprocessing involved one-hot encoding for categorical features and min-max normalization for numerical attributes. A small labeled portion was used for initial training, while the remaining data was streamed for inference. IDS-Twin was benchmarked against SOTA IDS baselines (AOC-IDS [8], CIDS [7], DT-IDS [9]) and standard ML models (RF, XGBoost, MLP), using accuracy, precision, recall, and F1-score as evaluation metrics. Key hyperparameters selected via grid search were $\tau = 0.07$, $m = 128$, $\lambda = 0.5$, $\alpha = 0.6$, $\beta = 0.4$, confidence threshold = 0.9, and similarity threshold $\delta = 0.3$, yielding an optimal trade-off between representation quality and detection performance.

DT Simulation Design. A key component of our experimental setup is a DT layer designed to emulate real-time behavior in a wireless IoT environment. The DT layer acts as a Python-based sequential replay engine, feeding flows one at a time in strict timestamp order to emulate real-time arrival. This emulates the online arrival of network flows, enabling IDS-Twin to operate under continuous monitoring and adaptive inference. Unlike traditional DTs that require physical-system coupling, IDS-Twin functions as a virtual DT by processing streamed flows that reflect the evolving system state. These flows are handled sequentially using a confidence-based mechanism that adaptively updates the model's decision boundaries. This software-defined approach mirrors the functional behavior of lightweight DT-based security frameworks, supporting dynamic model refinement without physical integration. IDS-Twin continuously updates itself during streaming through a feedback loop that combines historical and current observations. When 64 low-confidence flows accumulate, the system triggers online self-training by pseudo-labeling these flows and appending them to the training set. The DT replays flows in strict timestamp order, enabling incremental boundary refinement with asynchronous updates and providing controlled, reproducible evaluation using standard datasets without hardware.

Runtime profiling and efficiency evaluation. To assess IDS-Twin's suitability for real-time IoT deployments, we evaluated its responsiveness and adaptability under streaming conditions with high traffic volumes using both UNSW-NB15 and CIC-IoT-2023 datasets. The system processes each flow with an average inference time of approximately 6.8 ms, which includes both the encoder forward pass and the Gaussian-based confidence estimation. This enables timely decision-making in high-throughput environments. Additionally, the model

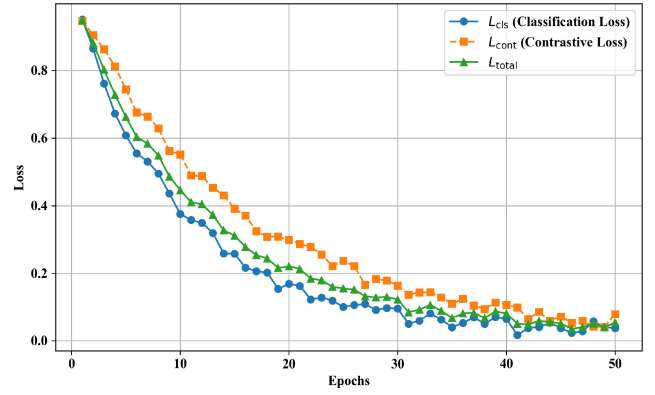


Fig. 2. Training loss convergence of IDS-Twin showing classification loss (L_{cls}), contrastive loss (L_{cont}), and total loss (L_{total}).

performs online self-training only when a buffer of 64 low-confidence flows is filled, reducing the frequency of model updates to approximately once every 800–1000 flows. This asynchronous adaptation mechanism minimizes runtime overhead while enabling continuous learning. Combined with its memory-efficient design that avoids storing the full traffic stream, IDS-Twin remains responsive and adaptive under large-scale network flow input. We also assessed the computational and memory overhead of IDS-Twin during deployment. The model maintains only a small adaptation buffer of 64 low-confidence samples, ensuring minimal memory consumption.

Training convergence analysis. Fig. 2 presents the training loss curves observed over the course of training, showing a consistent and steady decline in all components: classification loss L_{cls} , contrastive loss L_{cont} , and the combined objective L_{total} . The classification loss drops rapidly in the early epochs, indicating effective learning of class-discriminative features. In contrast, the contrastive loss decreases gradually, indicating the encoder's ability to learn consistent and well-separated representations.

B. Performance Comparison

Table I presents a comprehensive comparison of the proposed IDS-Twin framework against recent SOTA IDS baselines (AOC-IDS, CIDS, DT-IDS) and ML models (RF, XGBoost, MLP) on the UNSW-NB15 and CIC-IoT-2023 datasets. As shown, IDS-Twin consistently achieves the highest performance across all metrics. On the UNSW-NB15 dataset, IDS-Twin achieves an accuracy of 96.43%, outperforming AOC-IDS (93.45%) by 2.98%, CIDS (92.31%) by 4.12%, and DT-IDS (91.87%) by 4.56%. In terms of F1-score, IDS-Twin also leads with 95.77%, indicating a well-balanced trade-off between precision (95.66%) and recall (95.86%). This outperforms AOC-IDS (92.03%), CIDS (91.21%), and DT-IDS (90.27%), confirming IDS-Twin's superior capability in correctly identifying intrusions with fewer false alarms. On the more recent and heterogeneous CIC-IoT-2023 dataset, IDS-Twin again ranks highest, achieving 94.89% accuracy and a corresponding F1-score of 94.91%. Compared to AOC-IDS (91.70%) and CIDS (92.47%), IDS-Twin maintains a performance margin of at least 2.42% in accuracy and over 3% in F1-score. The recall score of

TABLE I
PERFORMANCE (%) COMPARISON BETWEEN IDS-TWIN AND BASELINES

Method	UNSW-NB15				CIC-IoT-2023			
	Acc.	Pre.	Rec.	F1	Acc.	Pre.	Rec.	F1
IDS-Twin	96.43	95.66	95.86	95.77	94.89	94.93	95.43	94.91
AOC-IDS	93.45	91.20	92.88	92.03	91.70	89.50	91.23	90.35
CIDS	92.31	90.22	91.87	91.21	92.47	90.51	91.43	90.94
DT-IDS	91.87	89.92	90.63	90.27	90.45	88.31	89.68	88.99
RF	89.19	88.56	89.55	89.01	90.74	88.98	89.71	89.43
XGBoost	91.56	90.12	91.77	90.93	89.82	90.47	91.16	90.80
MLP	90.80	91.40	92.10	91.76	91.11	91.66	92.30	91.96

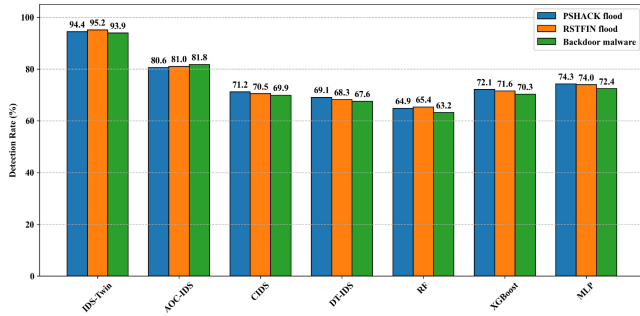


Fig. 3. Detection rates ($\uparrow$) of zero-day attacks for IDS-Twin and baselines.

95.43% also confirms IDS-Twin’s robustness in detecting diverse and potentially unseen IoT attack patterns. IDS-Twin surpasses baseline methods by combining contrastive representation learning, confidence-driven decision-making, and real-time self-supervised adaptation within a DT-based environment.

C. Zero-Day Attack Detection Analysis

We evaluate IDS-Twin’s generalization to zero-day attacks on CIC-IoT-2023 by withholding three attack types (*PSHACK flood*, *RSTFIN flood*, *Backdoor malware*) during training and introducing them only at test time, consistent with the theoretical definition in (2). Fig. 3 presents the detection rates achieved by IDS-Twin compared to six baseline methods, including recent SOTA baselines and traditional ML models. As shown, IDS-Twin consistently outperforms all baselines across all three zero-day attacks. It achieves detection rates of 94.43%, 95.17%, and 93.93% on *PSHACK flood*, *RSTFIN flood*, and *Backdoor malware*, respectively. In contrast, AOC-IDS and CIDS yield lower detection rates (around 80–82%), while DT-IDS (Stacked Ensemble) struggles to exceed 69%. RF, XGBoost, and MLP show further degradation, with detection rates falling between 63.2% and 74.3%. IDS-Twin achieves strong zero-day detection by generalizing to unseen attacks through contrastive representation learning, confidence-based uncertainty scoring, and adaptive pseudo-labeling in a DT.

V. CONCLUSION

This letter presents IDS-Twin, a DT-based intrusion detection framework for dynamic wireless IoT. By integrating

representation learning, confidence estimation, and online self-training, IDS-Twin addresses challenges of concept drift, scarce labels, and zero-day attacks. The DT environment supports real-time traffic replay and continuous feedback-driven adaptation. Experiments on benchmark datasets show that IDS-Twin achieves generalizable performance and consistently outperforms SOTA baselines, especially in zero-day scenarios. Future work will extend IDS-Twin with DT calibration, coupling to live telemetry, and integration into a complete detection-to-action workflow.

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